

Session 4.3

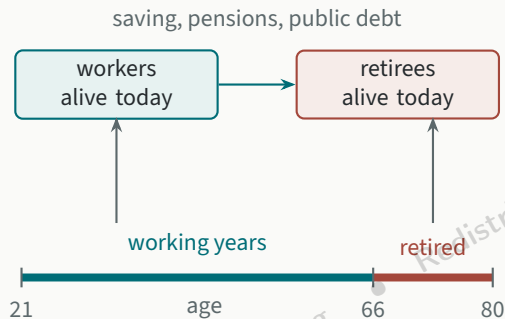
Cohorts Under Aggregate Risk

From Krusell–Smith to overlapping generations: a finite state, a network that clears a market, and why idiosyncratic risk sends us to histories



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From Krusell-Smith to Cohorts



- ▶ Deck 4.2: households live for ever; the state is $(a, \varepsilon; z, \Gamma)$ and Γ must be forecast.
- ▶ **Overlapping generations** (OLG): lives are finite, and cohorts of different ages coexist.
- ▶ Newborns hold nothing, the oldest leave, and Γ becomes a *stack*: one distribution for each age.
- ▶ **Same question as Deck 4.2, what stands for Γ , with a new structure to exploit.**

Source: Samuelson, *JPE* (1958); Diamond, *AER* (1965); Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987).

Two Kinds of Risk, Four Economies

	no aggregate risk	aggregate risk z_t
idiosyncratic ε	Part A: the classical life cycle; Γ a stack of cohort histograms; paths after one shock	Part C: every cohort a histogram, moving with z ; the state explodes
no ε	one household per cohort, no risk: a deterministic life cycle	Part B: one household per cohort; the state is 15 numbers; a network

- ▶ Parts A and B each drop one kind of risk, and each can then carry Γ exactly.
- ▶ **Part C puts both back, and borrows Deck 4.2's third method to survive it.**

Source: Krusell & Smith, *JPE* (1998); Ríos-Rull, *RES* (1996); Azinovic-Yang & Žemlička (2025).

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Cohorts Without Aggregate Risk

A household that ages, a distribution indexed by age, a path of prices



Finite Lives: The Last Period Is Known

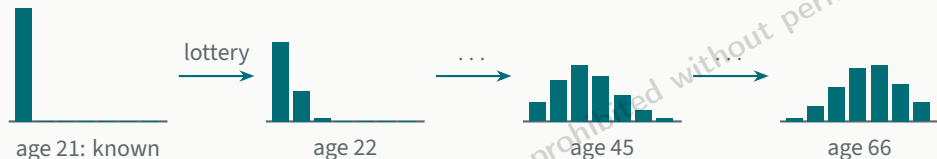


each arrow is one endogenous-grid step, using next year's policy

- ▶ The lab's life-cycle economy: 60 cohorts, ages 21 to 80, idiosyncratic ε , a limit $a' \geq 0$, a pension b .
- ▶ The policy depends on age: $a' = g_j(a, \varepsilon)$. At the last age consumption is cash on hand.
- ▶ Given c_{j+1} , Deck 4.1's Euler step delivers c_j : 59 steps in all, from age 80 back to 21. Deck 4.1's infinitely lived household instead iterates to a fixed point: 356 sweeps from a cold start ([the Session 4 lab, block A](#)).
- ▶ **Backward induction replaces iteration to convergence: the household problem has no fixed point left.**

Source: Auerbach & Kotlikoff (1987); Carroll, *Economics Letters* (2006); the Session 4 lab, block C2. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

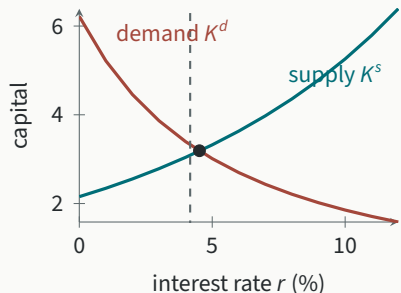
Cohorts Are the Distribution



- ▶ λ_1 is known: newborns hold nothing, with ε drawn from its invariant distribution.
- ▶ $\lambda_{j+1} = \lambda_j P_{g_j}$: Deck 4.1's *Young's Lottery*, applied once for each age.
- ▶ Given prices, no iteration to an invariant distribution: 59 pushes complete the cross-section. The price is still unknown: next page.
- ▶ **The economy's distribution is the stack of the cohorts'. Part C will need exactly this stack, moving with z .**

Source: Young, *JEDC* (2010); Ríos-Rull, *Review of Economic Studies* (1996).

The Stationary Equilibrium: One Price, Found by Bisection



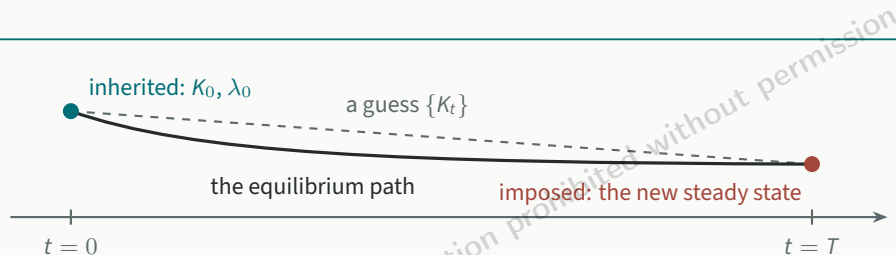
The stationary equilibrium

- 1 bracket $r \in [-7\%, 30\%]$; check the sign change
- 2 trial r : firm gives K/L , w ; tax pays pension
- 3 households backward: 59 EGM steps
- 4 cohorts forward: 59 pushes give $K^s(r)$
- 5 compare with $K^d(r)$; halve the bracket
- 6 stop after 29 halvings; residual 6×10^{-9}

- ▶ $r^* = 4.52\%$ lies above $1/\beta - 1 = 4.17\%$ (dashed): with finite lives that bound fails; the bracket must be wide.
- ▶ **One price, one fixed point; inside it, one backward and one forward pass.** Code: [the Session 4 lab, block C2](#).

Source: The Session 4 lab, block C2: 60 cohorts, a 40% pay-as-you-go pension.

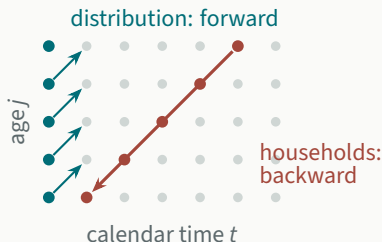
The Unknown Is a Path of Prices



- ▶ An unanticipated, permanent change at $t = 0$: an “MIT shock”. Afterwards the path is known.
- ▶ Unknown: $\{K_t\}_{t=0}^T$. Prices $\{r_t, w_t\}$ follow from the firm’s conditions.
- ▶ Boundary conditions at both ends: λ_0 is inherited, and behaviour at T is the new steady state’s.
- ▶ **Decisions look forward and distributions look back. No single pass can do both.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger, *RED* (1999).

Update, Damp, Repeat



The transition loop

- 1 solve both steady states; choose the horizon T
- 2 guess $\{K_t\}$; prices follow from it
- 3 households backward along each cohort's diagonal
- 4 distribution forward from λ_0 : the supply K_t^s
- 5 stop when $\max_t |K_t^s - K_t|/K_t$ is small
- 6 else $K_t \leftarrow (1 - d)K_t + dK_t^s, K_0$ fixed

- Damping $d = 0.5$: a higher guess lowers r_t and the saving meant to support it, so a full step can oscillate.
- [the Session 4 lab, block C](#) runs this loop for Deck 4.1's infinitely lived economy after a 20% tax on capital income: 91 passes to 1×10^{-7} , $T = 300$. **The outer loop is a fixed point in a sequence.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger (1999); the Session 4 lab, block C. Annual: $\alpha = 0.33, \beta = 0.96, \delta = 0.08, \sigma = 2, \rho_\varepsilon = 0.90, a' \geq 0$.

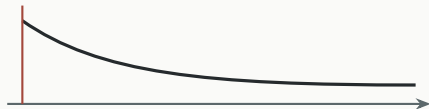
Measuring Who Gains

- ▶ **Consumption-equivalent variation** $g(a, \varepsilon)$: scale the old consumption plan by $(1 + g)$ at every date and in every state, until its value matches the new one. $g > 0$ is a gain.
- ▶ With CRRA utility V scales by $(1 + g)^{1-\sigma}$, so $g(a, \varepsilon) = [V_1(a, \varepsilon)/V_0(a, \varepsilon)]^{1/(1-\sigma)} - 1$.
- ▶ Everything turns on *which* V_1 . Along the path: the value at $t = 0$ of living through the transition. Between steady states: the value of waking up in the new one.
- ▶ Average with the distribution of the people who are asked: λ_0 , the households alive at the reform.
- ▶ **One formula, two numbers. The difference is whether the path is counted.**

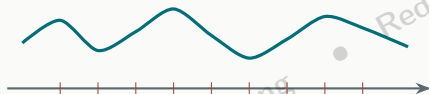
Source: Lucas, *Models of Business Cycles* (1987); Domeij & Heathcote, *International Economic Review* (2004).

One Shock Is Not a Business Cycle

one shock



Part A: a known path afterwards



a shock every period: no path to compute

- ▶ A path answers one unanticipated shock, after which everyone knows the future.
- ▶ With z_t drawn every period, households must forecast; the policy is again a function of the state (z, Γ) .
- ▶ That is Krusell and Smith's problem, now with a stack of cohort distributions.
- ▶ **Part B: drop idiosyncratic risk, and the stack becomes a short vector a network can read.**

Source: Krusell & Smith (1998); Boppart, Krusell & Mitman, *JEDC* (2018).

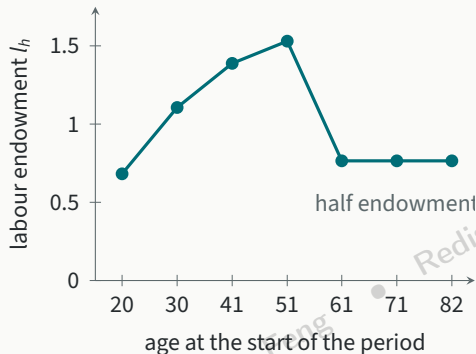
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Aggregate Risk Without Idiosyncratic Risk

After Azinovic-Yang & Žemlička, *Economics-Inspired Neural Networks with Stabilizing Homotopies*: without ϵ a cohort is one household, and the distribution is a vector



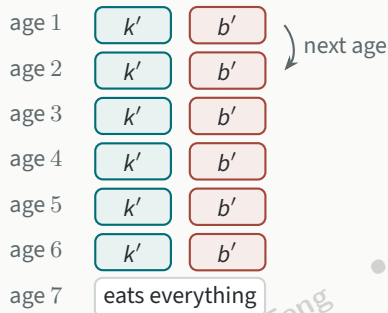
The Economy: Seven Cohorts, Two Assets



- ▶ $H = 7$ cohorts; one period is $72/7 \approx 10.3$ years, ages 20 to 82. Endowments l_h (left), summing to 7.
- ▶ CRRA $\gamma = 2$; $\beta = 0.96^{72/7}$. Two assets: capital k and a one-period bond b .
- ▶ TFP: $\log z' = \rho \log z + \sigma \epsilon$, annual $\rho = 0.85$, $\sigma = 0.03$, rescaled to the period.
- ▶ Depreciation moves with TFP: $\delta_t = \min\{\delta \frac{2}{1+z_t}, 1\}$, δ from 10% a year. **A bad draw also destroys capital.**

Source: Part B follows Azinovic-Yang & Žemlička, “Economics-Inspired Neural Networks with Stabilizing Homotopies”, arXiv 2303.14802, and its OLG notebook (*L_profile*, *Sigma_normalize*, the calibration cells). In PyTorch, scaled for a CPU: [the Session 4 lab, block F](#).

The Household: Capital, Bonds, Adjustment Costs



- ▶ Age h with holdings (k, b) chooses (k', b') :
$$c + p b' + k' + \psi_k (k' - k)^2 = l_h w + b + (1 - \delta + r) k.$$
- ▶ $k' \geq 0$; $b' \geq b_{\min} = -0.05$; the last cohort saves nothing. $\psi_k = 0.1$: changing one's own capital is costly.
- ▶ Six ages choose: 6 + 6 saving decisions.
- ▶ **No ε : everyone of an age is identical, so a cohort is one household.**

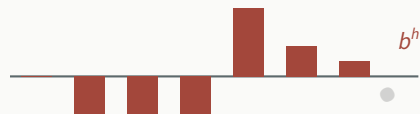
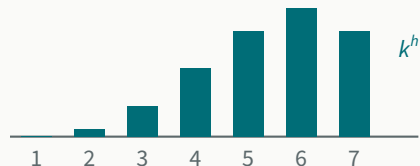
Source: Notebook markdown “Households” and the budget in E ; the constraint cell.

Firm and Markets

- ▶ Firm: $Y = z K^\alpha L^{1-\alpha}$, $\alpha = 1/3$; $r = \alpha Y/K$, $w = (1 - \alpha)Y/L$.
- ▶ Aggregates: $K = \sum_h k^h$, $L = \sum_h l_h = 7$, fixed.
- ▶ Bonds in zero net supply: $\sum_h b'^h = B = 0$. The bond price p is the one price not given by a marginal product.
- ▶ Goods clear by Walras' law.
- ▶ The notebook's text swaps the formulas for r and w ; the code, `get_firm_side`, is as written here.
- ▶ **Two assets, one of them priced by the market: a network must now return a price as well as policies.**

Source: Notebook `get_firm_side`; market-clearing markdown.

The State Is Fifteen Numbers

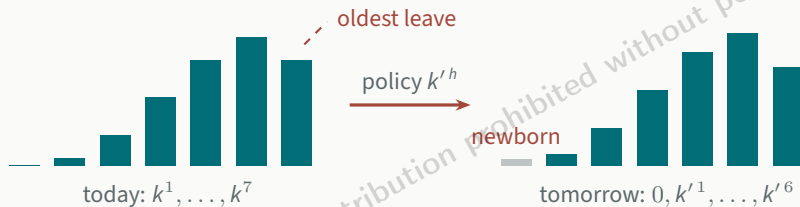


age h ; bars schematic

- ▶ $x = (z, k^1, \dots, k^7, b^1, \dots, b^7) \in \mathbb{R}^{15}$. Newborns hold nothing; the code keeps $k^1 = b^1 = 0$ anyway.
- ▶ Unknowns: 6 capital policies, 6 bond policies and the bond price, as functions of x : 13 outputs.
- ▶ Equilibrium: $12 = 6 \times 2$ KKT conditions (six ages that save, two assets each) and bond clearing, at every x .
- ▶ **Unlike Deck 4.2: Γ is exact. No moment, no histogram, no history.**

Source: Notebook “State Variable and Equilibrium Functions”; index constants for the state.

Tomorrow's Distribution Is a Shift



- ▶ Age h 's choice today is age $h + 1$'s holding tomorrow; a newborn enters with $(0, 0)$.
- ▶ The only randomness in x' is z' : one successor for each quadrature node.
- ▶ **Unlike Deck 4.2's lottery: no mass to split between grid points. The policy's output is the next state.**

Source: Notebook *ForwardStep* and the “next period wake-up assets” in E .

Equilibrium Conditions as Fischer–Burmeister Residuals

- ▶ KKT conditions for age $h < 7$, written as relative consumption errors, u'^{-1} the inverse marginal utility:

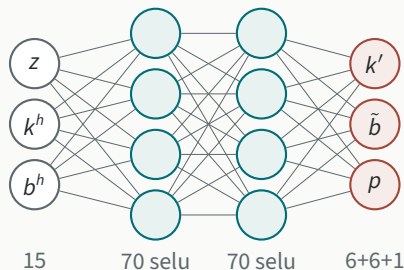
$$\varepsilon_k = \psi^{FB} \left(\frac{u'^{-1}(\beta \mathbb{E}[u'(c') (1 - \delta' + r' + 2\psi_k(k'' - k'))]) / (1 + 2\psi_k(k' - k))}{c} - 1, \frac{k'}{c} \right),$$

$$\varepsilon_b = \psi^{FB} \left(\frac{u'^{-1}(\beta \mathbb{E}[u'(c')]) / p}{c} - 1, \frac{b' - b_{\min}}{c} \right), \quad \psi^{FB}(a, b) = a + b - \sqrt{a^2 + b^2}.$$

- ▶ $\psi^{FB} = 0$ exactly when $a \geq 0, b \geq 0, ab = 0$: the Euler equation holds, or the limit binds.
- ▶ $\mathbb{E}$: 8-node Gauss–Hermite quadrature over z' . c' and k'' from the same network at x' .
- ▶ **As Deck 4.2's Deep Equilibrium Nets, one period ahead; now two assets and twelve residuals per state.**

Source: Notebook *E* (capital and bond terms), *fischer_burmeister*; Maliar, Maliar & Winant, *JME* (2021).

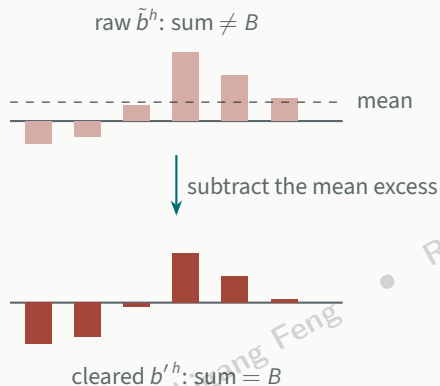
The Network: Fifteen Numbers In, Thirteen Out



- ▶ One network for all 13 functions: shared hidden layers, so ages share what they learn.
- ▶ Capital: $k' = 0.1 \text{ softplus}(o_k) \geq 0$, so no short sales by construction.
- ▶ Price: $p = \text{softplus}(o_p) > 0$.
- ▶ Bonds: the raw $\tilde{b}$ goes through a clearing layer (next page).
- ▶ **Consumption is not an output: it is backed out of the budget.**

Source: Notebook *base_net_construct*, *S_fun*; hidden width $N = 10 \times 7$.

A Layer That Clears the Bond Market



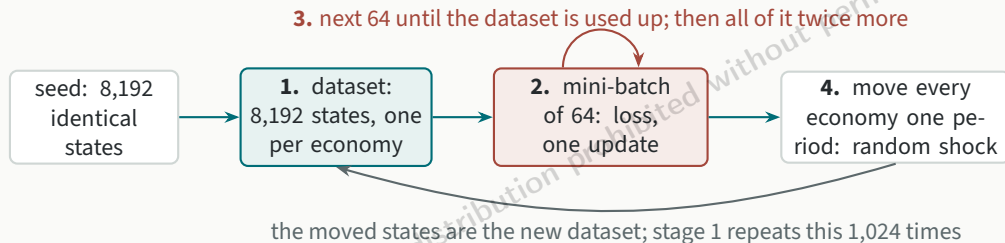
- From the six raw outputs:

$$b'^h = m \tilde{b}^h - \frac{\sum_j m \tilde{b}^j - B}{6}.$$

- Then $\sum_h b'^h = B$ holds *exactly*, at every state and every step of training.
- m is a mask the homotopy uses: $m = 0$ removes bonds altogether.
- **Unlike all three methods of Deck 4.2: a price is an output, and a market clears by construction, not by a loss term.**

Source: Notebook *S_fun*; the paper (market-clearing layers).

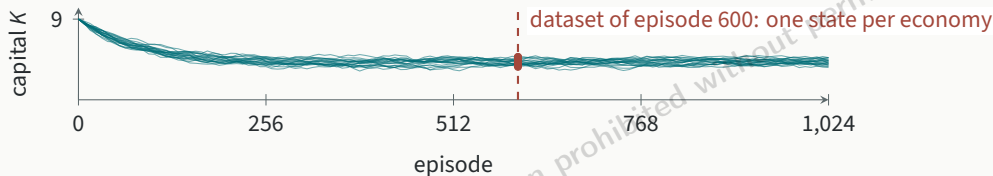
How the Network Is Trained: the Big Picture



- ▶ **1. Samples:** 8,192 economies simulated side by side; their current states are the dataset.
- ▶ **2. Mini-batch:** 64 states, the loss at them, one gradient step.
- ▶ **3. Epoch:** 128 mini-batches use the dataset once; 3 epochs per episode.
- ▶ **4. Episode:** every economy moves one period; stage 1 repeats this 1,024 times.

Source: Notebook *CloudMethod*, *Episode*, *TrainingLoop*, *ForwardStep*; settings *n_cloud*, *n_minib*, *epochs_epis*.

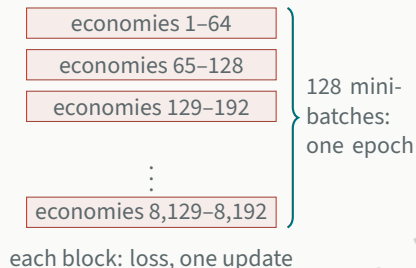
1. How the Samples Are Generated



- ▶ Each line is one economy's capital; 20 of the 8,192 are drawn. 8,192 is a setting: the number of economies simulated side by side.
- ▶ All start identical: $z = 1, k^{2..7} = 1.5$ ($K = 9$), no bonds. They drift apart only because each draws its own shocks.
- ▶ The dataset of an episode is the vertical slice: 8,192 states, one per economy, each $x = (z, k^{1..7}, b^{1..7})$.
- ▶ **Not a stored panel: only the current slice is kept, and it is replaced each episode.**

Source: Notebook settings ($n_{cloud} = 2^{13}$), the seed (*representative_input*). Paths schematic.

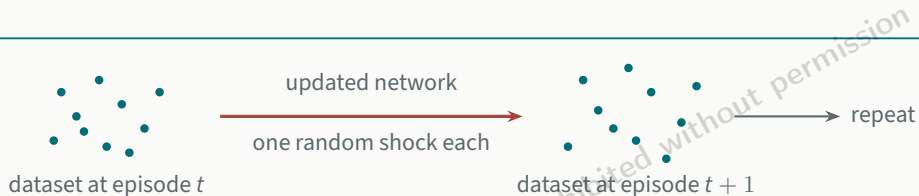
2-3. Mini-Batches, and an Epoch



- ▶ A mini-batch: 64 states taken *in order*, economies 1-64, then 65-128, and so on (64 is a setting).
- ▶ Each: the loss at its 64 states, one gradient step (Adam, 10^{-5}).
- ▶ $8,192/64 = 128$ mini-batches use the dataset up: one epoch.
- ▶ 3 epochs on the same dataset, the same 128 mini-batches each time: $3 \times 128 = 384$ updates; each state enters the loss 3 times.

Source: Notebook *TrainingLoop* ($n_{\text{minib}} = 2^6$, $\text{epochs_epis} = 3$), *Update*.

4. Episodes: the Dataset Moves One Period



- ▶ After the 3 epochs, each economy moves one period: its policy from the updated network, its own shock drawn at 2σ (the loss uses σ , so the data reach past the economy's usual range).
- ▶ The moved states replace the old ones and are the next episode's dataset.
- ▶ Stage 1, the capital-only economy, runs 1,024 episodes; the later homotopy stages continue from the same 8,192 economies.
- ▶ **This is the only simulation step. The look-ahead inside the loss (next page) computes an expectation and moves nothing.**

Source: Notebook *Episode*, *ForwardStep*, *CloudMethod*; *shock_mult* = 2 (Bretscher et al., 2022).

5. Inside One Update: One Mini-Batch, Line by Line

One update on a mini-batch of 64 states

- 1 take the next 64 states x of the dataset, in order
 - 2 for each: $K = \sum_h k^h$; r, w from z, K ; one call gives $k', \tilde{b} \rightarrow b'$ (clearing layer), p ; c from the budget
 - 3 build 8 successors $x'_j = (\rho \log z + \sqrt{2}\sigma\eta_j, 0, k'^{1..6}, 0, b'^{1..6})$, one per quadrature node η_j for z'
 - 4 call the network at each x'_j : k'', b'', p' ; then c'_j from the budget at $t + 1$
 - 5 $\mathbb{E}[\cdot] = \sum_j \omega_j(\cdot)$; the residuals $\varepsilon_k, \varepsilon_b$ for ages 1–6
 - 6 mean loss over the 64; gradient through all 9 calls, k', b' inside x'_j too; Adam, 10^{-5}
- Steps 3–5 are the look-ahead: 8 fixed quadrature nodes for z' , used only to form the expectation. They do not move the dataset. Per state, $1 + 8 = 9$ network calls; per update, $9 \times 64 = 576$.
 - **Unlike Deck 4.2's Method 2, whose successors are stored data: here they are rebuilt inside the loss, so the gradient sees the move to x' .**

Source: Notebook *E, vE, Loss, GradAndLoss, Update*.

The Loss on One Mini-Batch

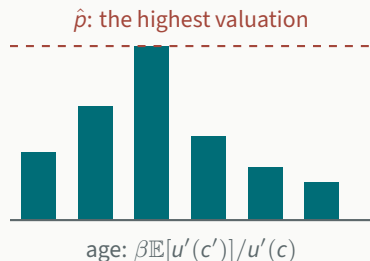
- ▶ A mini-batch B is 64 states x (the mini-batch size, 2^6 , a setting); residuals are averaged over states and ages:

$$\mathcal{L}(\theta; B) = \frac{1}{2} \overline{\varepsilon_k^2} + \frac{1}{2} w_b \overline{\varepsilon_b^2} + w_{\text{pre}} \overline{\varepsilon_{\text{pre}}^2} + 100 \overline{\varepsilon_c^2}.$$

- ▶ $\varepsilon_c = \max(c, 10^{-4}) - c$: a network that implies negative consumption is pushed back hard.
- ▶ $\varepsilon_{\text{pre}} = p - \text{sg}(\hat{p})$: a target price, with its gradient stopped.
- ▶ w_b and w_{pre} are switched by the homotopy. One Adam step per mini-batch.
- ▶ **Unlike Deck 4.2's methods, the loss weights change during training: the loss is a path of problems.**

Source: Notebook *Loss, GradAndLoss, E (eps_cnn, eps_bpr)*.

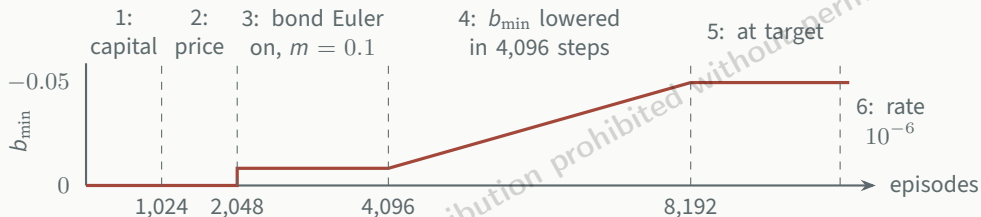
Why a Homotopy: a Bond Nobody Can Trade Has No Price



- ▶ At $b_{\min} = 0$ and $B = 0$ no one can hold a bond, so any high enough price clears the market: p is not pinned down.
- ▶ Its natural limit: the **zero-liquidity price** $\hat{p} = \max_h \beta \mathbb{E}[u'(c'^{h+1})]/u'(c^h)$, what the most eager saver would pay for the first bond.
- ▶ Then relax $b_{\min}$ a little at a time; each problem starts from the last one's solution.
- ▶ **Solving the full model from a random start is where state-space training tends to fail.**

Source: Notebook “Pretrain bond price”, $p_{b_implied}$; the paper.

The Homotopy in Six Stages



- ▶ 1: $w_b = w_{\text{pre}} = m = 0$, 1,024 episodes. 2: $w_{\text{pre}} = 1$, learn $\hat{p}$, 1,024. 3: $w_b = 1$, $w_{\text{pre}} = 0$, $m = 0.1$, $b_{\min} = -0.05/6$, 2,048.
- ▶ 4: one episode per step down to -0.05 . 5: 2,024 there. 6: 124 at a tenth of the rate. Each inherits cloud, weights, optimiser.
- ▶ **Each stage starts from the last one's network and cloud: the same 8,192 tracks run through all six, 10,340 periods.**

Source: Notebook, "Initialize Capital-Only Model" to fine-tuning; a comment says mask 0.01, the code sets 0.1.

Ten Numbers, and What Each One Is

number	what it counts	why
7	cohorts, each 72/7 years	a life cycle short enough to write out
15 / 13	network inputs / outputs	the exact state; 12 policies and a price
70	units in each of 2 hidden layers	$10 \times$ the number of cohorts
8,192	states in the cloud	wide batches instead of long simulations
64 / 384	mini-batch / Adam steps per episode	$3 \times 8,192/64$
3	passes over each cloud	then it moves one period
8	Gauss-Hermite nodes for z'	successors per state, inside the loss
2	multiple of σ when simulating	spread the training states
4,096	steps lowering $b_{\min}$ to -0.05	one episode each
$10^{-5} / 10^{-6}$	Adam rate / in the last 124 episodes	small and steady

Source: Notebook “Algorithm Choices” and the homotopy cells.

Deep Equilibrium Nets on the OLG, as an Algorithm

The homotopy notebook

- 1 seed a cloud of 8,192 identical states; a random network
- 2 for each stage (capital; price; bonds on; lower $b_{\min}$; target; fine-tune):
 - 3 set the homotopy: $b_{\min}$, mask m , weights w_b, w_{pre}
 - 4 for each episode of the stage:
 - 5 384 Adam steps: 64 states, 9 network calls each, FB residuals
 - 6 move the whole cloud one period with fresh shocks at 2σ
 - ▶ About 10,300 episodes, or four million Adam steps; the notebook times each stage in minutes on one GPU.
 - ▶ **The same loop as Deck 4.2's Deep Equilibrium Nets, with three changes: an exact state, a clearing layer, and a path of problems.**

Source: Notebook *CloudMethod*, *Episode*, and the homotopy loop.

Four Methods, One Table

	DeepHAM	Deep Eq. Nets, KS	Sequence space	Deep Eq. Nets, OLG (here)
what stands for Γ	learned moments of 50 households	two 100-point histograms	the last 50 shocks	Γ itself: 14 numbers
what is minimised	minus lifetime utility; a value regression	Euler and KKT residuals	an FB Euler residual	FB residuals; bond clearing built in
where states come from	panels, refreshed	16 economies, carried forward	a replay buffer	a cloud of 8,192, moved one period
what the gradient passes through	150 periods of the panel	one period, at stored states	one period; prices fixed	one period, through the computed x'

► **Row 1 is why the OLG is easy here: with no ε , nothing has to stand for Γ .**

Source: Deck 4.2, *Four Questions Every Deep Method Answers*; the homotopy notebook.

What Changed From Deck 4.2

- ▶ **An exact state.** Cohorts without ε are points, so the distribution is a vector, and its law of motion is a shift.
- ▶ **A price as an output.** Two assets need a bond price; the network returns it, and a layer clears the market exactly.
- ▶ **A path of problems.** Capital first, then a price, then bonds that trade a little more each step.
- ▶ **Wide and shallow simulation.** 8,192 economies moved one period at a time, at twice the shock's s.d.
- ▶ **Successors differentiated.** The loss builds x' from the policy; the gradient sees the move.
- ▶ **All five rest on the first. Take it away and the rest has nothing to stand on.**

Source: The homotopy notebook; Deck 4.2.

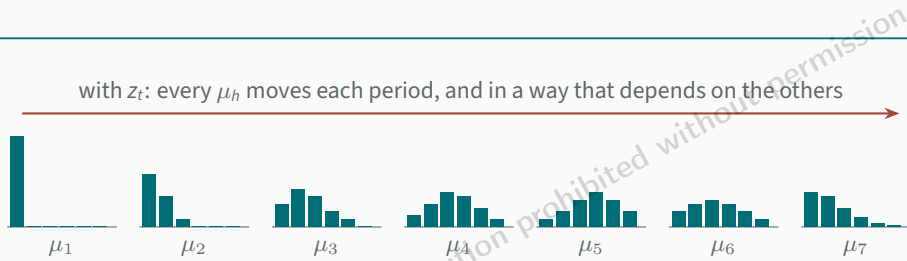
Session 4.3

Put Idiosyncratic Risk Back

After Azinovic-Yang & Žemlička, *Deep Learning in the Sequence Space*: every cohort becomes a distribution, and the state explodes



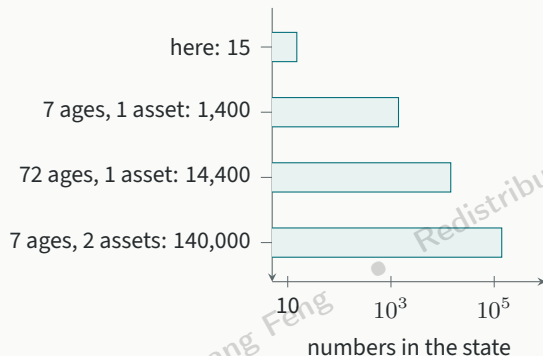
Each Cohort Becomes a Histogram



- ▶ With ε , two households of the same age differ. A cohort is a distribution $\mu_h(\varepsilon, k)$ again, as in Part A.
- ▶ The aggregate state: $(z, \mu_1, \dots, \mu_H)$. Part A's stack, but no longer stationary: it moves with z .
- ▶ **Part A had the stack without z ; Part B had z without the stack. Now both.**

Source: Part C follows Azinovic-Yang & Žemlička, “Deep Learning in the Sequence Space”, working paper, arXiv 2509.13623 (2025), §6, with their slides (2026); here p. 113.

Count the Inputs



Source: Arithmetic. Cohort count: their slides, p. 13.

- ▶ Two income states, 100 asset points per cohort: $7 \times 2 \times 100 = 1,400$.
- ▶ With the published OLG's 72 cohorts: $72 \times 2 \times 100 = 14,400$.
- ▶ Two assets on a 100×100 grid: $7 \times 2 \times 10^4 = 140,000$.
- ▶ **A state-space network reads all of it, and its training states move whenever the policy does.**

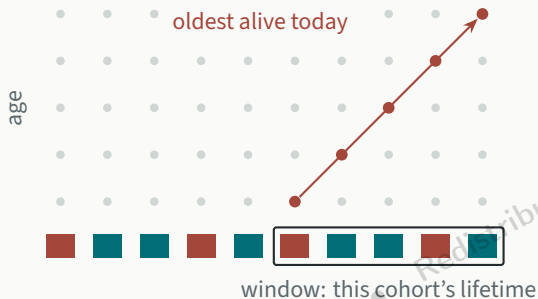
Why Deck 4.2's First Two Answers Strain

	Krusell–Smith, one distribution	OLG, H distributions
classical rule	forecast K' from K	forecast every cohort's moments, or prices from many of them
DeepHAM	one panel of 50, one learned moment	a panel for each age; moments per age
Deep Eq. Nets	2×100 masses as input	$H \times 2 \times 100$ masses, drifting with the policy
sequence space	the last 50 shocks	the same window of shocks

- ▶ The first three grow with the number of ages; the last does not, because its input never contained Γ .
- ▶ Deck 4.2, *What the Exogenous Domain Buys*: large inputs that shift with the policy feed the unstable loop.
- ▶ **Only the history keeps its size when cohorts are added.**

Source: Deck 4.2; their slides, pp. 11, 23–24.

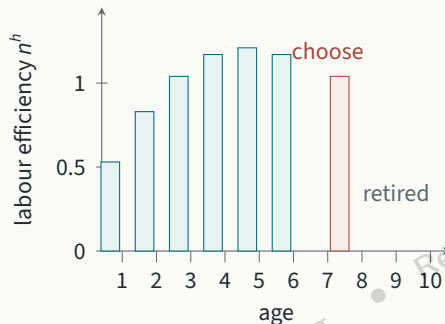
The Way Out: the History Already Holds Every Cohort



- ▶ Everyone alive was born within the last H periods: μ_h is built by the shocks of its lifetime, from its starting wealth.
- ▶ Older shocks act only through prices and inheritances (bequests, in the model ahead): Deck 4.2's **memory loss**.
- ▶ So the stack is, approximately, a function of recent shocks: a window at least a lifetime long.
- ▶ **Published sweep, 1 to 256 lags: accuracy improves with the window.**

Source: Intuition ours; evidence: their slides, pp. 7–8, 20–21, 91 (the paper's 72-cohort OLG).

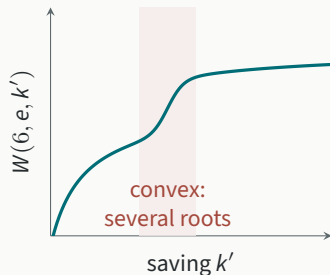
The Economy: Ten Cohorts, Income Risk, a Retirement Choice



- ▶ 10 cohorts of six years: ages 1–6 work, age 7 chooses to work or retire (logit, taste-shock scale 0.08), ages 8–10 are retired. Bequests become newborns' assets.
- ▶ Income $e \in \{0.5, 1, 1.5\}$, Markov; one asset, capital, $k \geq 0$.
- ▶ Aggregate risk: three regimes z ; the third is a disaster, depreciation 0.397 instead of 0.265.
- ▶ State (z, μ) , $\mu(h, e, k)$ on $10 \times 3 \times 300$: 9,000 masses. Input: the last 100 regimes.
- ▶ **The network never reads μ .**

Source: The paper, §6.1–6.2; their slides, pp. 105–113. The paper numbers ages 0–9; its implementation text says 200 asset points, its parameter table 300.

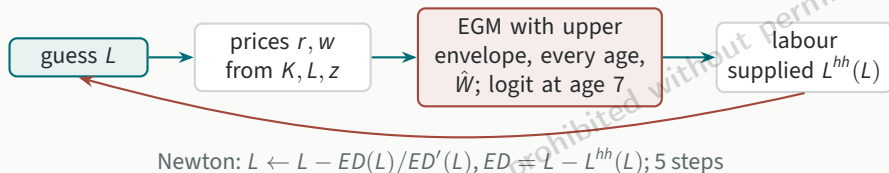
Why the Network Outputs Continuation Values



- ▶ Age 6 looks ahead to retirement, so its value of saving is convex in places (left, schematic): the Euler equation has several roots, and its residual can pick the wrong one.
- ▶ So the network outputs **continuation values** $\hat{W}(h, e, k') \approx \mathbb{E}[V(h+1, e', k')]$ on the grid, ages 1–9; EGM with an **upper envelope** keeps the best root.
- ▶ Increasing by construction: a first value plus softplus increments. One call gives all $9 \times 3 \times 300$: operator learning.
- ▶ **Unlike Deck 4.2's tutorial: no consumption output, no FB residual.**

Source: The paper, §6.3; their slides, pp. 110, 114–120; Fella (2014); Druedahl & Jørgensen (2017).

Within the Period: From $\hat{W}$ to Policies, and a Labour Market That Must Clear



- ▶ K is fixed by μ ; L is not: whether age 7 works depends on the wage, and the wage on L .
- ▶ Five Newton steps, ED' by automatic differentiation, from $L^{\text{work}} + \hat{\lambda} L^{7, \text{max}}$: $\hat{\lambda}$ is the network's guess of the share of age 7's labour supplied.
- ▶ Result: prices, every age's saving rule $g^h(e, k)$, the work probabilities $p^{\text{work}}(e, k)$, and the realised share λ .
- ▶ **Unlike Part B, where a layer cleared the bond market exactly: here a price is found by iteration inside every evaluation.**

Source: The paper, §6.3 (Step 0 of the forward simulation).

Moving the Distribution One Period

- ▶ Young's lottery with income transitions, for a saving rule g and masses $m(i, j)$ at (e_i, k_j) :

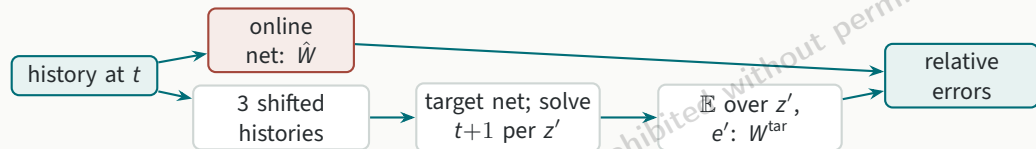
$$[\mathcal{T}(g, m)](i', j') = \sum_{i, j} m(i, j) \Pi_{ii'}^e (\omega^- \mathbf{1}\{j' = j^-\} + \omega^+ \mathbf{1}\{j' = j^+\}),$$

$j^\pm$ the grid points around $g(e_i, k_j)$, $\omega^\pm$ the lottery weights of Deck 4.2.

- ▶ Aging: $\mu'_{h+1} = \mathcal{T}(g^h, \mu_h)$. Age 7 splits: $\mathcal{T}(g^{\text{work}}, p^{\text{work}} \odot \mu_7) + \mathcal{T}(g^{\text{ret}}, (1 - p^{\text{work}}) \odot \mu_7)$.
- ▶ Newborns: $\mu'_1 = \mathcal{T}(g^{10}, \mu_{10})$, the bequests.
- ▶ μ' does not depend on z' : three candidate successors (z', μ') , each with its history shifted by z' .
- ▶ **The same lottery as Deck 4.2's Methods 2 and 3, applied age by age.**

Source: The paper, §6.3 (Steps 1–3 of the forward simulation); Young, *JEDC* (2010).

The Loss: Targets From a Slower Copy of the Network



- ▶ Target: $W^{\text{tar}}(h, e, k') = \sum_{z'} \Pi_{zz'}^z \sum_{e'} \Pi_{ee'}^e V_{t+1}(h+1, e', k' \mid z'; \bar{\rho})$, $t+1$ solved with the target network $\bar{\rho}$.
- ▶ Loss: mean squared relative errors, $(\text{target} - \text{prediction}) / (\epsilon_{\text{loss}} + |\text{target}|)$, on W at the first grid point, on W 's increments, and on λ ; ϵ_{loss} from 1 to 0.1 to 0.01.
- ▶ Targets fixed; gradient through the online network only; then $\bar{\rho} \leftarrow \tau \rho + (1 - \tau) \bar{\rho}$, $\tau = 10^{-4}$.
- ▶ **Unlike the tutorial's residual: a supervised target from a slower copy, as Deck 4.1's critic.**

Source: The paper, §6.3 (loss function) and §6.4 (Stage 3).

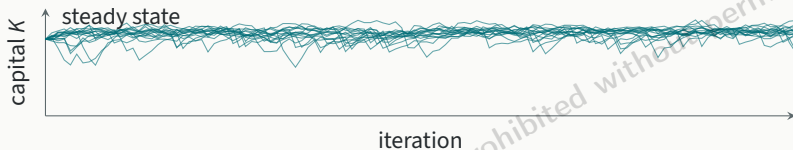
One Iteration, Line by Line

One Adam step on 1,024 pairs (z, μ , history)

- 1 online network at each history: $\hat{W}$ and $\hat{\lambda}$
 - 2 target network at the 3 shifted histories
 - 3 clear labour at t (5 Newton steps, each an EGM solve): prices, policies, λ
 - 4 move μ one period: 3 candidate successors
 - 5 solve $t+1$ for each z' with the target network; take the expectation: W^{tar}
 - 6 loss; one Adam step at 10^{-6} ; update the target network
 - 7 draw z' from Π^z ; that successor is the pair's next state
- ▶ Three stages: (1) a deterministic steady state by EGM, which seeds the pairs; (2) aggregate risk on, continuation values held at the steady state, to spread the pairs; (3) the loop above.
 - ▶ Network: three hidden layers of 1,024 (GELU); history length 100.

Source: The paper, §6.4 (outline, Stages 1–3, hyperparameters).

Where the 1,024 Pairs Come From, and What They Cover



- ▶ Start: the steady-state distribution. Each iteration moves every pair one period: z' from Π^z , μ' by the lottery, the history shifted. Dips: disaster regimes.
- ▶ As in Part B, many tracks, one period per step; unlike Deck 4.2's Method 3, no pool and no branching.
- ▶ Coverage: histories are drawn from Π^z , so the 3^{100} windows are met at their true frequencies; a life is 10 periods, the window 100. μ comes along with its history.
- ▶ **Coverage is checked on simulated paths, not guaranteed.**

Source: The paper, §6.3 (Step 3) and §6.4 (step 7). Paths schematic.

Bridge: Each Tool for Its Question

Paths: one unanticipated shock, any distribution.

The state as input: when Γ is a short vector, as here.

The history as input: when Γ is many distributions.

Next: back to Deck 4.1's histogram, and a government that taxes on it.

- ▶ Age turns Γ into a stack. Without ε the stack is a vector, and Deep Equilibrium Nets solve it exactly.
- ▶ With ε the stack is many histograms: the history is the input, continuation values the output, EGM the solver.
- ▶ **Deck 4.4: a government that cannot commit chooses a tax rule on the whole distribution.**

Source: The sources of this deck.